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# Enhancing SIEM Systems Using Time Correlation

## Graduation Project 2

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## List of Abbreviations and Technical Terms

| Abbreviation | Meaning |
|---|---|
| SIEM | Security Information and Event Management |
| LANL | Los Alamos National Laboratory |
| AUTH | Authentication event records |
| PROC | Process event records |
| DNS | Domain Name System |
| FLOWS | Network flow records |
| AI | Artificial Intelligence |
| PoC | Proof of Concept |
| GNN | Graph Neural Network |
| EPS | Events Per Second |
| RAM | Random Access Memory |
| CPU | Central Processing Unit |
| Wazuh | Open-source security monitoring / SIEM platform |
| Meta-event | A summarized unit linked to its original contributing events |
| Provenance | Traceable origin and supporting evidence of an event |

## Abstract

Security Information and Event Management (SIEM) systems receive large, heterogeneous streams of security events. Processing every record independently may increase operational cost, while excessive reduction can hide temporal sequences, original evidence, and relationships among devices. This project investigates whether evidence-preserving, graph-based temporal aggregation can reduce the workload of a SIEM without unacceptably weakening security detection or cross-device incident correlation.

The proposed research extends an earlier temporal pre-correlation prototype. It examines synchronized authentication, process, DNS, and network-flow records from the Los Alamos National Laboratory (LANL) cyber-security dataset. The design consists of three connected research functions: constructing a temporal graph that retains traceable event references; developing device-level Attack/Normal assessment within time windows; and combining graph evidence with model outputs to distinguish shared attacks, independent attacks, and unresolved relationships. Graph-based aggregation is additionally investigated to produce traceable meta-events for controlled Wazuh workload testing. The final graph schema, aggregation algorithm, AI model, and integration arrangement remain subject to experimental comparison.

The study will compare temporal graph representations and graph-based aggregation strategies while preserving original event references. Subsequent evaluation will examine the volume of SIEM-facing meta-events, evidence integrity, computational overhead, device-level attack assessment, and cross-device incident correlation. These are research objectives and planned evaluation dimensions; no performance improvement is assumed before controlled experiments.

Keywords: SIEM, temporal correlation, graph-based aggregation, LANL, event provenance, attack detection, cross-device correlation, Wazuh.

## الملخص

باستخدام الارتباط الزمني، مع التركيز على (SIEM) يتناول مشروع تحسين أنظمة إدارة المعلومات والأحداث الأمنية تخفيض عبء معالجة السجلات الأمنية دون فقدان الأدلة المهمة للكشف عن الهجمات. تنشأ المشكلة من ضخامة الأحداث وتنوّع مصادرها، ومن احتمال فقدان الترتيب الزمني والعلاقات بين الأجهزة عند دمج السجلات أو اختصارها بطريقة غير مدروسة.

يقترح المشروع دراسة تمثيل الأحداث ضمن رسم بياني زمني مع الحفاظ على معرّفات الأحداث ومراجعها الأصلية، ثم تطوير نموذج ذكاء اصطناعي لتقييم نشاط كل جهاز ضمن نافذة زمنية، وبعد ذلك الاستفادة من نتائج النموذج وأدلة الرسم البياني لتحديد ما إذا كانت الأجهزة المشبوهة تشترك في هجوم واحد أو ترتبط بهجمات مستقلة أو لا تتوافر أدلة كافية للحسم. كما ويعتمد Wazuh. تُدرس آليات تجميع مبنية على الرسم البياني لإنتاج أحداث ملخّصة قابلة للتتبع، تمهيدًا لقياس أثرها في منصة LANL. من بيانات FLOWSو DNSو PROCو AUTH العمل مبدئيًا على مصادر

تقوم منهجية البحث على مقارنة تصاميم مختلفة للرسم البياني الزمني وطرق تجميع الأحداث مع المحافظة على إمكانية استرجاع السجلات الأصلية. وتشمل خطة التقييم قياس عبء المعالجة وجودة الاحتفاظ بالأدلة، ودراسة كشف النشاط الهجومي على مستوى الجهاز، ثم اختبار الربط بين الأجهزة. وتُعامل الفوائد المتوقعة بوصفها فرضيات بحثية تحتاج إلى تجارب موثقة، لا نتائج مثبتة مسبقًا.



الكلمات المفتاحية: SIEM، الارتباط الزمني، Graph Aggregation، LANL، كشف الهجمات، حفظ الأدلة، Wazuh.

# Chapter 1: Introduction

## 1.1 Chapter Introduction

Security Information and Event Management (SIEM) platforms bring together authentication logs, process activity, DNS observations, network-flow records, and other security information. Correlation across these sources can help security analysts identify suspicious sequences that would be difficult to recognize in isolated records. Yet high event volume creates a tension: systems must process security data efficiently without discarding the evidence needed to understand an incident [1], [2].

Graduation Project 1 addressed part of this problem through temporal pre-correlation on a principal LANL AUTH path. Graduation Project 2 extends that foundation by investigating whether a temporal graph can preserve relationships across multiple event sources, support meaningful event aggregation, and assist subsequent device-level detection and incident correlation. The project's final improvement claims will depend on measured experiments rather than assumptions about graph compression.

## 1.2 Project Overview

The official project title remains "Enhancing SIEM systems using time correlation." Its central research aim is to optimize SIEM event processing using evidence-preserving graph-based temporal aggregation while maintaining useful information for security analysis. The project is designed as an experimental, locally executed prototype rather than an industrial production platform.

The data path begins by inspecting source-specific LANL fields, synchronizing selected time intervals, and assigning traceable references to original records. A temporal heterogeneous graph can represent event nodes and observed host, account, or process-name associations. Graph aggregation methods will then be evaluated for their ability to create fewer meta-events while retaining original timestamps, membership, and provenance references.

A connected but distinct function analyzes whether each device exhibits attack-related activity within a time window. An independent graph-correlation function then uses the model outputs and observed relationships to determine whether suspicious devices belong to the same incident, are demonstrably involved in independent attacks, or require an Unknown conclusion. Wazuh is the planned SIEM platform for evaluating the effect of aggregated outputs and overall computational overhead. The exact integration order is not yet finalized.

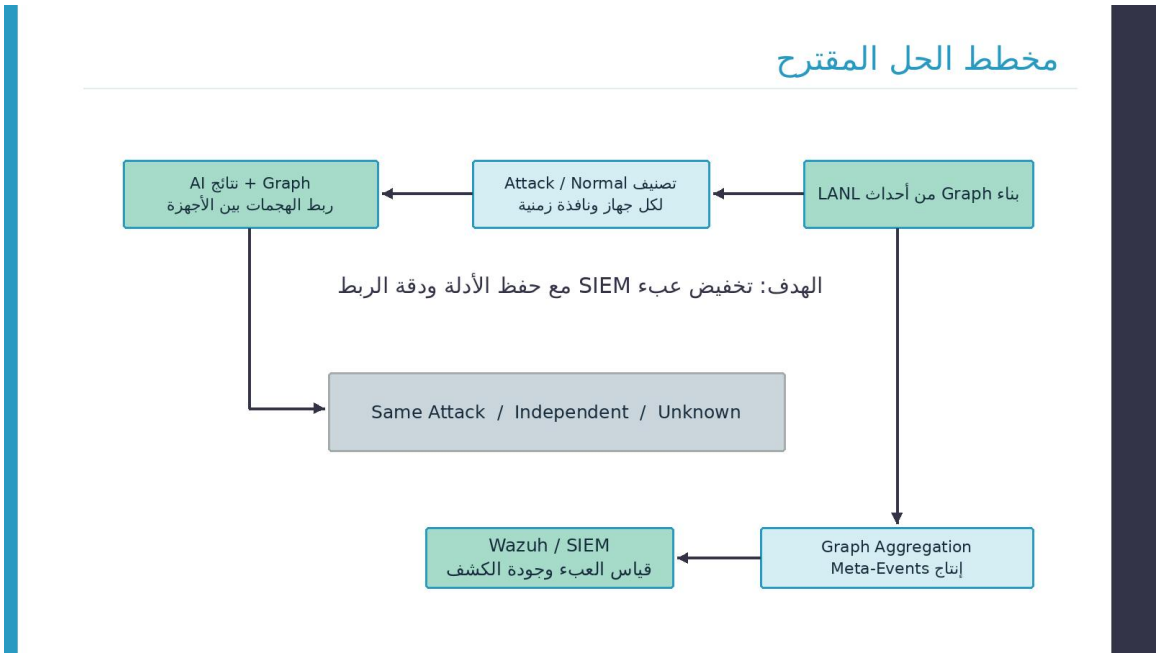



*Figure 1.1 — Proposed solution overview for Graduation Project 2.*

### 1.3 Scientific Problem and Project Justification

The problem is not merely that SIEM systems receive many logs. Different sources record different portions of a potentially related sequence of actions, and many records may look repetitive without being interchangeable. Discarding or merging them solely on superficial similarity can erase timing, device identity, or important evidence. Conversely, preserving every record without useful organization may increase ingestion, storage, correlation, and investigation cost [1]–[3].

A second challenge is cross-device interpretation. Two computers may be suspicious during the same interval, but temporal overlap by itself does not demonstrate that they are participating in one coordinated attack. Reliable incident association requires observed evidence, well-defined inference rules, and the possibility of withholding a conclusion when the available dataset is insufficient [4], [5].

The research gap examined here is therefore an integrated evaluation question: can a multi-source temporal graph support actual SIEM-facing event reduction, maintain auditable source evidence, and still provide useful context for device-level detection and cross-device attack grouping, while its full processing cost is measured? Prior studies contribute graph compression, alert grouping, anomaly detection, and SIEM evaluation separately, but their published metrics cannot be treated as an end-to-end result for this specific LANL–Wazuh setup [1]–[5].

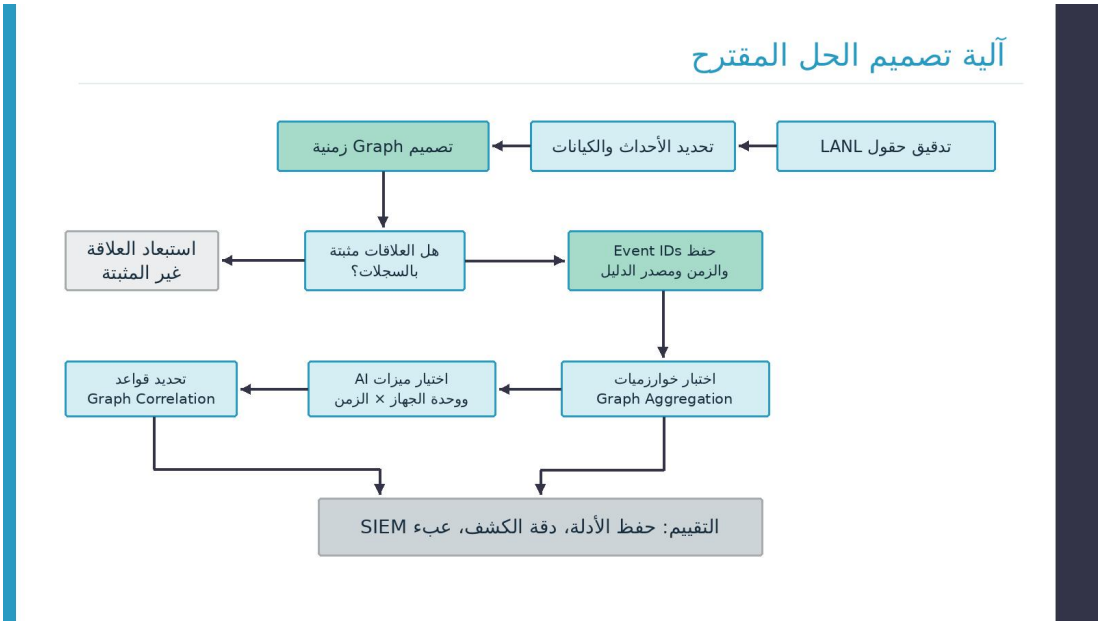



*Figure 1.2 — Design mechanism of the proposed solution.*

## 1.4 Project Objectives

The general objective is to investigate an efficient and auditable time-aware event-processing approach that reduces SIEM workload without unacceptable loss of relevant security evidence or detection utility.

- Inspect and align the original fields and timestamps of LANL AUTH, PROC, DNS, and FLOWS records.
- Construct and compare evidence-preserving temporal graph representations without inventing causal or process-parent relationships.
- Evaluate graph-based grouping algorithms that emit traceable meta-events rather than merely reducing an internal graph representation.
- Develop and validate a device-by-time-window Attack/Normal assessment approach under clearly stated labeling limitations.
- Correlate detected device activity using graph evidence to assess Same Attack, Independent Attacks, or Unknown.
- Compare raw input, the Graduation Project 1 temporal baseline, and graph-based approaches under matched data and evaluation settings.
- Measure event and byte reduction, evidence integrity, graph time/memory, detection quality where valid labels exist, and Wazuh and end-to-end computational cost.

### 1.5 Research Questions

- Can traceable graph-based aggregation reduce the number and size of records delivered to SIEM without losing critical original-event evidence?
- Which temporal graph representation and event-grouping constraints offer the best measured balance between overhead and context preservation on LANL?
- How reliably can suspicious activity be detected for individual devices within time windows when labels are incomplete?
- Can the combination of graph evidence and device-level detection distinguish shared attacks from independent incidents without relying on temporal coincidence alone?
- Does any reduction of Wazuh workload remain beneficial after graph construction, aggregation, AI, and incident-correlation costs are included?

### 1.6 Importance of the Project

Scientifically, the work examines the trade-off between information reduction and evidence preservation, and separates three tasks often conflated in security analytics: representing or compacting event relationships, detecting suspicious behavior, and determining whether suspicious devices belong to a common incident. This separation makes claims easier to test.

Practically, the work may identify ways to reduce unnecessary SIEM processing while retaining event references needed for investigation. Its experimental design also addresses an important limitation: reducing the size of a graph or the number of alerts reviewed by an analyst does not automatically establish lower ingestion cost or higher detection accuracy [1], [2], [6].

Methodologically, the project emphasizes matched baselines, reproducible event identifiers, temporal boundaries, and explicit handling of missing ground truth. The value of the approach will ultimately be determined by the measured outcome and its limitations.

### 1.7 Related Studies Supporting the Project

The following four open-access or freely accessible original studies are selected as central references for the initial research narrative. Other supporting literature remains available in the project's Literature Evidence Matrix v3 and will be used in later chapters when justified. These studies address complementary questions and are not directly comparable by a single accuracy or compression metric.

| Study / Year | Publication | Reported contribution | Relevance to this project |
|---|---|---|---|
| CSCProv (2026) [1] | Cybersecurity | Causal-semantic constraints for compressing provenance graphs while retaining attack | Motivates evidence-preserving grouping and auditable graph reduction. |

| | | | |
|---|---|---|---|
| | | context. | |
| From Logs to Tactics (2026) [2] | International Journal of Information Security | Produces meta-alerts and reconstructs multi-stage attack campaigns. | Supports the distinction between event grouping and incident-level correlation. |
| KAIROS (2024) [4] | IEEE Symposium on Security and Privacy | Detects anomalous activity from provenance and reconstructs attack context. | Informs the separate detection and investigation functions. |
| Cybersecurity on a Budget (2024) [6] | PLOS ONE | Benchmarks open-source SIEM tools, including Wazuh. | Guides operational evaluation; not proof of our system performance. |

ProvGRP [3] additionally motivates behavior-aware graph reduction, and graph-based alert contextualization [5] supports studying event-centric representations. These works inform candidate experiments but do not establish the final graph schema or algorithm.

### 1.8 Current State of the Art in the Project Area

Research on security event analysis includes several related but distinct approaches. Provenance-graph methods can compact or analyze graphs while attempting to retain meaningful dependencies [1], [3]. Alert-grouping methods form higher-level analytical units and may reconstruct campaigns from correlated observations [2]. Temporal and graph-based detection systems identify suspicious events or sequences and support post-detection investigation [4], [5]. SIEM benchmarking, meanwhile, measures the performance of deployed tools under controlled workloads [6].

These directions provide the methodological foundation for our research, but they cannot be transplanted without adjustment. In particular, detailed causal provenance datasets may include process identifiers, file operations, and socket relationships not present in the chosen LANL sources. Thus, claims about causal graph reconstruction must not be made from simple time proximity, shared usernames, or repeated host names alone.

### 1.9 Challenges Addressed by the Project

- Heterogeneous source schemas and uncertain alignment of identifiers across logs.
- Preservation of original Event IDs, time ordering, and source evidence after meta-event formation.
- Risk of misleading graph edges or clustering caused by high-degree hosts and accounts.
- Incomplete labels in LANL Redteam data; absence of a redteam record is not a Normal label.
- Difficulty of establishing authoritative Same/Independent incident labels for cross-device evaluation.
- Limited 16 GB local RAM and the combined resource demands of graph processing, AI, and Wazuh.

- Need to compare workload reduction and total end-to-end cost rather than isolated compression ratios.

### 1.10 Practical Applications of the Project

Potential applications include SIEM log pre-processing, creation of traceable higher-level event summaries, enrichment of analyst investigation context, and experimental correlation of security activity across devices. A successful prototype could inform practical ingestion strategies or security data-engineering pipelines. However, the project does not currently claim production readiness, automated incident response, or generalization to arbitrary enterprise datasets.

### 1.11 General Scope of the Project

The study focuses on the LANL cyber1 dataset, using AUTH, PROC, DNS, and FLOWS as the intended multi-source inputs. Work is organized around data inspection, temporal graph representation, evidence-preserving event aggregation, device-level security assessment, cross-device incident correlation, and evaluation with Wazuh. All graph designs, aggregation methods, and model configurations remain subject to systematic validation and comparison.

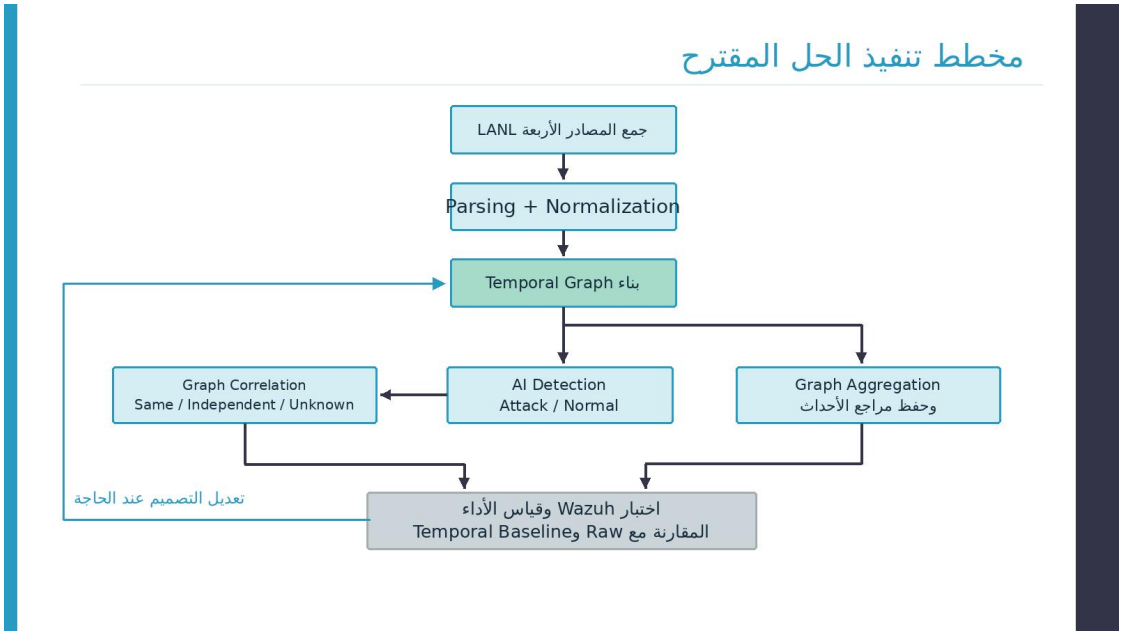



*Figure 1.3 — Execution flow of the proposed solution.*

The Hybrid Temporal Graph is an initial candidate and will be compared with event-centric and entity-centric alternatives. The historical Graduation Project 1 baseline uses a 5-second temporal aggregation configuration on its AUTH main path; this is a reference setting rather than a fixed window for every component in Graduation Project 2. AI modeling, actual meta-event

generation, cross-device incident decisions, and Wazuh benchmarking remain planned experiments at the current documented stage.

Project work is staged across literature review, data preparation, graph representation, graph-based grouping, device-level attack detection, graph-based incident correlation, Wazuh-oriented evaluation, report writing, and seminar preparation. These activities may overlap as experimental findings refine later design choices and documentation.

### 1.12 Thesis Structure

This partial document includes the preliminary matter and Chapter 1 only, following the structure used in the Graduation Project 1 final report and the department reporting guide. In the intended complete report, Chapter 2 will present the theoretical framework and research foundations; Chapter 3 will describe the work environment, data preparation, and methodology; Chapter 4 will describe the proposed system and implementation; Chapter 5 will present experiments, comparisons, and validated findings; and the conclusion will summarize achieved results, limitations, and future directions. Those later chapters are not drafted or included in this version.

### 1.13 Chapter Conclusion

This chapter introduced the scientific problem, research motivation, objectives, questions, related studies, and scope of Graduation Project 2. The central question is whether evidence-preserving temporal graph methods can reduce the number of events processed by a SIEM while retaining the context necessary for device-level detection and cross-device incident correlation. The effectiveness and operating cost of the proposed approach will be determined by subsequent controlled experiments.

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